Parameters and architecture

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable architecture and parameter-count record.

Count slots,sort quantities and explain why size is not measured quality.

Keep one habit: W1–W3/W5 → selected new architecture → project definition → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Forward passes by hand. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Representing **input**s as vectors. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Consider a dense network with 2 **input**s, 3 hidden units using ReLU, and 1 **output** unit using identity. We call its widths 2 to 3 to 1. These widths, connectivity and activation choices describe its **architecture**.

  The weights and biases supply the numerical values used by that **architecture**.

- A fully connected hidden unit receiving 2 **input**s needs 2 weights plus 1 bias. Three such units need 6 weights and 3 biases. The **output** receives 3 activations and needs 3 weights plus 1 bias. Total: 13 **parameter**s.

  Input values and computed activations are not additional **parameter**s.

- This counting rule assumes one separate bias per **output** unit and no shared or tied weights. Other **architecture**s can use different rules.

Forward passes by hand

> **My remembered idea, example and correction**

Activation functions

My remembered idea, example and correction

Representing inputs as vectors

My remembered idea, example and correction

Lesson reference for workbook practice

Layer	Weights	Biases	Total
2 inputs to 3 hidden units	2 times 3 = 6	3	9
3 hidden values to 1 output	3 times 1 = 3	1	4
Whole network	9	4	13

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Fully connected 2→3→1, hiddenReLU/outputidentity, one separate bias per receiving unit, no sharing. Hidden6 weights+3 biases=9; output 3 weights+1 bias=4, total 13. Dense n→mcountnm+m. Hidden width 4 gives12+5=17, +4 from2 incomingweights/1 hiddenbias/1 outgoingweight. No-bias variant 6+3=9. Activation.7is computed, weight−2a parameter, width 3an architecture choice, learningrate.1a training setting.

  ReLU has no fitted values here. Helper assumes positive integer widths. Count alone establishes neither same model nor accuracy, resource use or generalization. Independent3→2→2 supplies a new architecture to count.

- A **dense layer** has 4 **input**s and 2 **output**s with one bias per **output**. Count weights, biases and total.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Fully connected 2→3→1, hiddenReLU/outputidentity, one separate bias per receiving unit, no sharing. Hidden6 weights+3 biases=9; output 3 weights+1 bias=4, total 13. Dense n→mcountnm+m. Hidden width 4 gives12+5=17, +4 from2 incomingweights/1 hiddenbias/1 outgoingweight. No-bias variant 6+3=9. Activation.7is computed, weight−2a parameter, width 3an architecture choice, learningrate.1a training setting.

  ReLU has no fitted values here. Helper assumes positive integer widths. Count alone establishes neither same model nor accuracy, resource use or generalization. Independent3→2→2 supplies a new architecture to count.

- Classify hidden activation 0.7, weight -2, hidden width 3 and learning rate 0.1 as computed value, **parameter**, **architecture** choice or **training** setting.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Fully connected 2→3→1, hiddenReLU/outputidentity, one separate bias per receiving unit, no sharing. Hidden6 weights+3 biases=9; output 3 weights+1 bias=4, total 13. Dense n→mcountnm+m. Hidden width 4 gives12+5=17, +4 from2 incomingweights/1 hiddenbias/1 outgoingweight. No-bias variant 6+3=9. Activation.7is computed, weight–2a parameter, width 3an architecture choice, learningrate.1a training setting.

  ReLU has no fitted values here. Helper assumes positive integer widths. Count alone establishes neither same model nor accuracy, resource use or generalization. Independent3→2→2 supplies a new architecture to count.

- Recount the 2 to 3 to 1 model if every bias is omitted. State what the original count assumed.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Fully connected 2→3→1, hiddenReLU/outputidentity, one separate bias per receiving unit, no sharing. Hidden6 weights+3 biases=9; output 3 weights+1 bias=4, total 13. Dense n→mcountnm+m. Hidden width 4 gives12+5=17, +4 from2 incomingweights/1 hiddenbias/1 outgoingweight. No-bias variant 6+3=9. Activation.7is computed, weight−2a parameter, width 3an architecture choice, learningrate.1a training setting.

  ReLU has no fitted values here. Helper assumes positive integer widths. Count alone establishes neither same model nor accuracy, resource use or generalization. Independent3→2→2 supplies a new architecture to count.

- A team says its 26-**parameter** model must outperform a 14-**parameter** model. Give a fair check and one reason the claim is unsupported.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A dense 3 to 2 to 2 network uses ReLU in the hidden layer and identity in the **output** layer, with one bias per unit. Give each layer count and the total.

  If hidden width becomes 4, recount and explain why activation values are excluded.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Count a new architecture independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled **parameter**s and **architecture evidence** sheet to Image-pattern Network Investigation. Show how you can distinguish architectural choices from **parameter** values.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Sort the quantities for **Parameter**s and **architecture**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Parameter**s and **architecture**. Proposed explanation: "Count one bias per **input**". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A dense 3 to 2 to 2 network uses ReLU in the hidden layer and identity in the **output** layer, with one bias per unit. Give each layer count and the total.

  If hidden width becomes 4, recount and explain why activation values are excluded.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

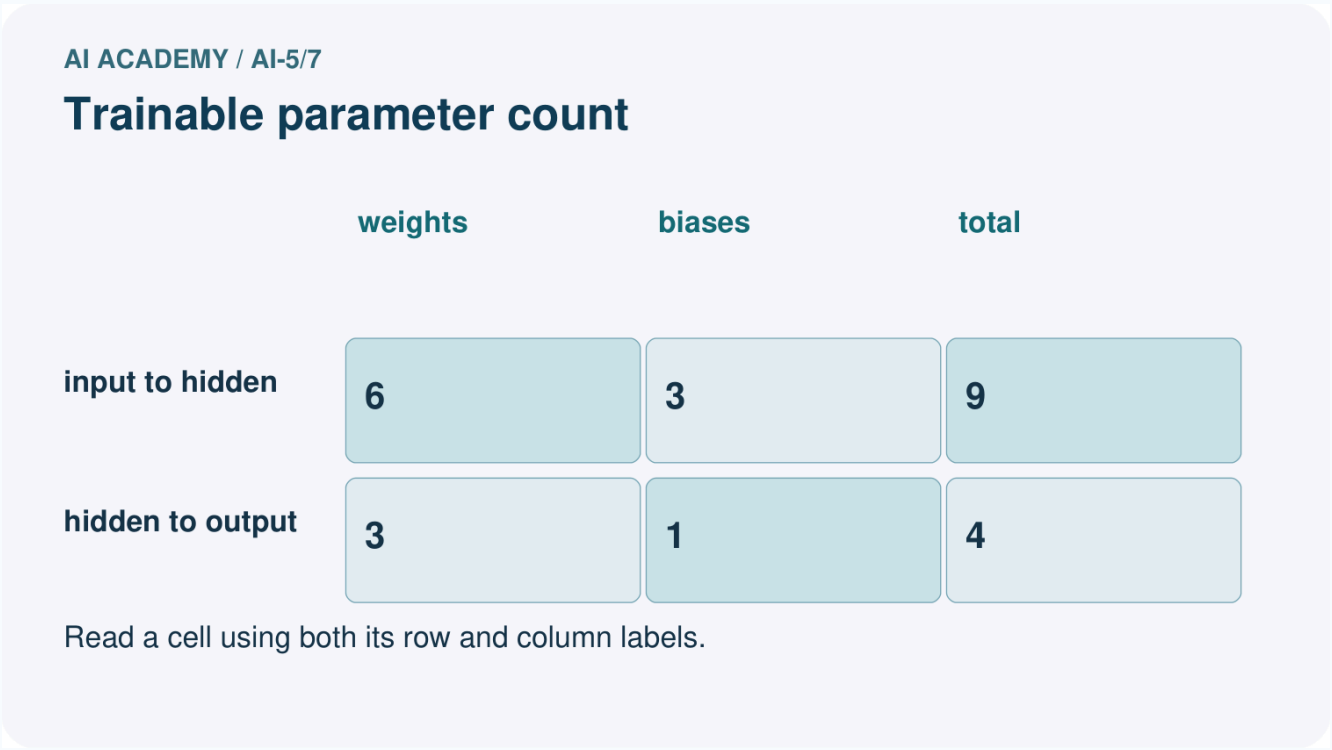
> Keep this task's inputs,rule and response together.

Complete task and evidence

- Count trainable weights and biases separately from activation functions.
- Dense toy **architecture** 2 **input**s→3 hidden units→1 **output**. Every hidden and **output** unit has one bias; hidden activation is ReLU.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

- How many new **parameter**s did that one hidden unit add? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/7

Parameters and architecture

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count first layer

2 x 3=6 weights plus 3 biases gives 9 parameters.

↓



2. Count output layer

3 x 1=3 weights plus 1 bias gives 4 parameters.

↓



3. Total

Total trainable parameters=9+4=13; ReLU adds no parameters here.

CHANGE ONE THING

Increase hidden width from 3 to4.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Count trainable weights and biases separately from activation functions.
- Which labelled step matches this detail? Total trainable **parameter**s=9+4=13; ReLU adds no **parameter**s here.
- 1. Total
- 2. Count **output** layer
- 3. Count first layer
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Total Total trainable **parameter**s=9+4=13; ReLU adds no **parameter**s here. Count one weight for each **input**-to-unit connection and one bias for each receiving unit.

Two **input**s connecting to three hidden units make six weights. The three hidden **output**s connect to one **output** unit through three more weights. ReLU applies a fixed function and adds no adjustable **parameter** here.

Architecture determines the connection pattern; learned values occupy those connections.

Changed-case question and optional hint

- Changed-case practice: Increase hidden width from 3 to4. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. First layer 12 **parameter**s; **output** layer 5; total 17.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: First layer 12 **parameter**s; **output** layer 5; total 17. Each new hidden unit adds 2 **input** weights, 1 hidden bias and 1 outgoing weight.

Repair and retain the record

Review Forward passes by hand (ai-5/6). Goal: Record scores and activations in dependency order. Short practice: Why should feedback identify h before correcting only y?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Activation functions (ai-5/4). Goal: Trace identity ReLU and sigmoid on a numerical score. Short practice: Which original **input**s cannot be distinguished by this network's final **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Representing **input**s as vectors (ai-5/1). Goal: Encode a record using a fixed feature order and units. Short practice: Why does equal total brightness not make A and B identical **input**s?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Parameters and architecture: the case

AI ACADEMY / AI-5/7 · 01 · LOOK AT THE FACTS

Parameters and architecture: the case

The situation

Count trainable weights and biases separately from activation functions.

Supplied evidence

Dense toy architecture 2 inputs→3 hidden units→1 output. Every hidden and output unit has one bias; hidden activation is ReLU.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain parameters and architecture using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

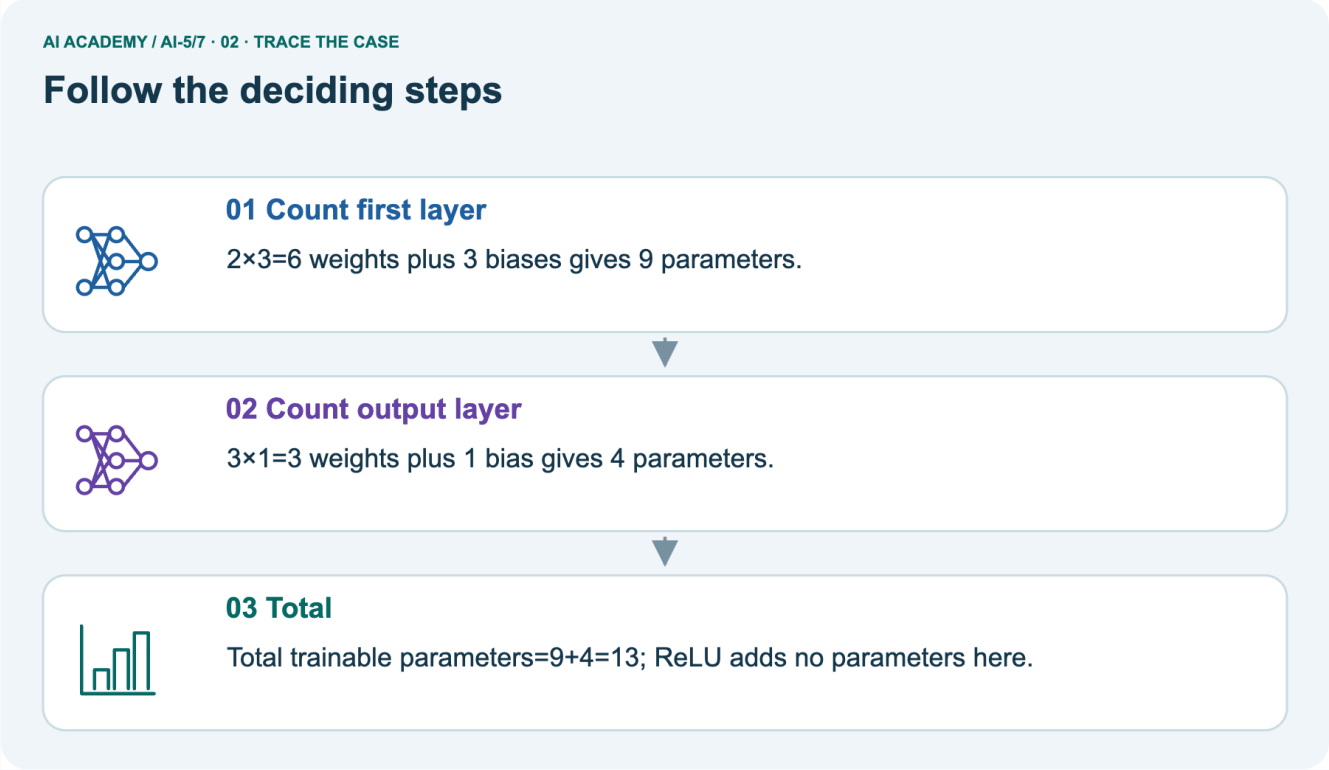
Count trainable weights and biases separately from activation functions. Supplied facts: Dense toy **architecture** 2 **input**s→3 hidden units→1 **output**. Every hidden and **output** unit has one bias; hidden activation is ReLU.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



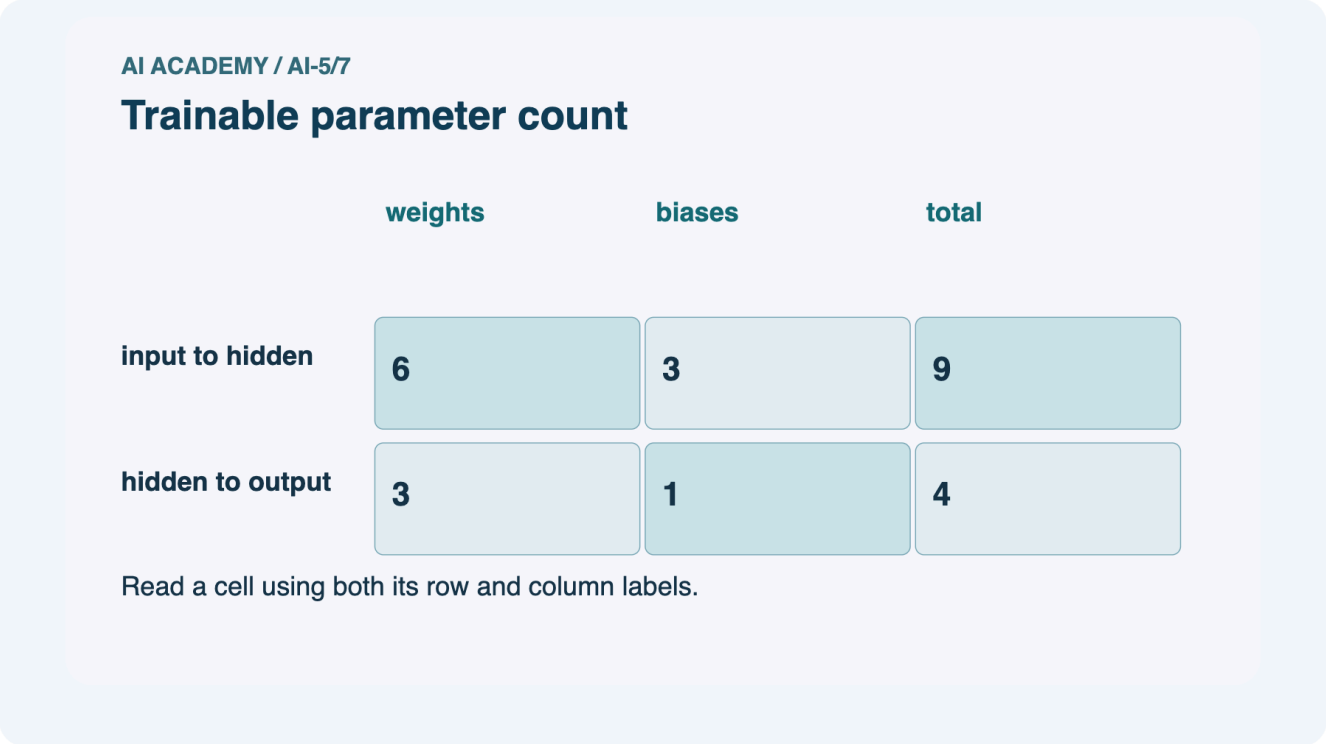
Follow the labelled picture with the matching explanation. Open the vector diagram

Count first layer: 2×3=6 weights plus 3 biases gives 9 **parameter**s. → Count **output** layer: 3×1=3 weights plus 1 bias gives 4 **parameter**s. → Total: Total trainable **parameter**s=9+4=13; ReLU adds no **parameter**s here.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Trainable parameter count



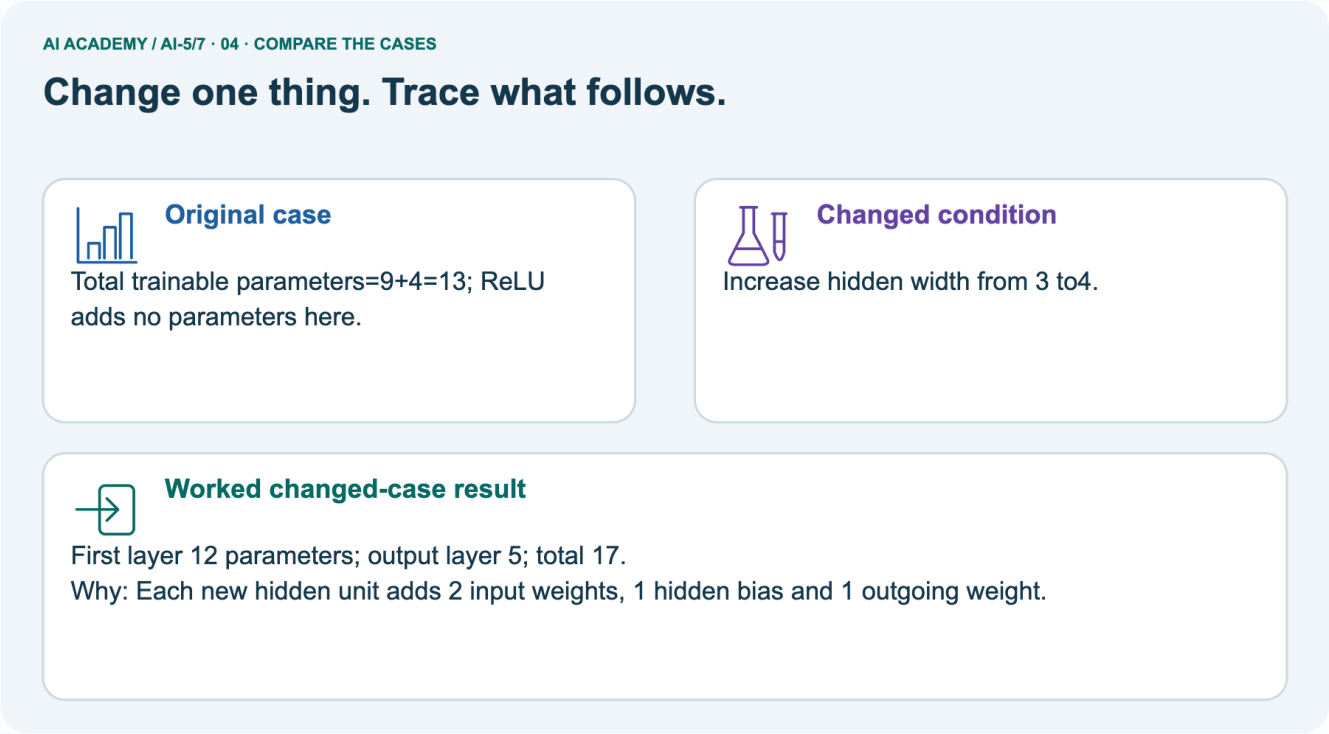
Follow the labelled picture with the matching explanation. Open the vector diagram

Trainable **parameter** count. **input** to hidden: weights = 6, biases = 3, total = 9. hidden to **output**: weights = 3, biases = 1, total = 4.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

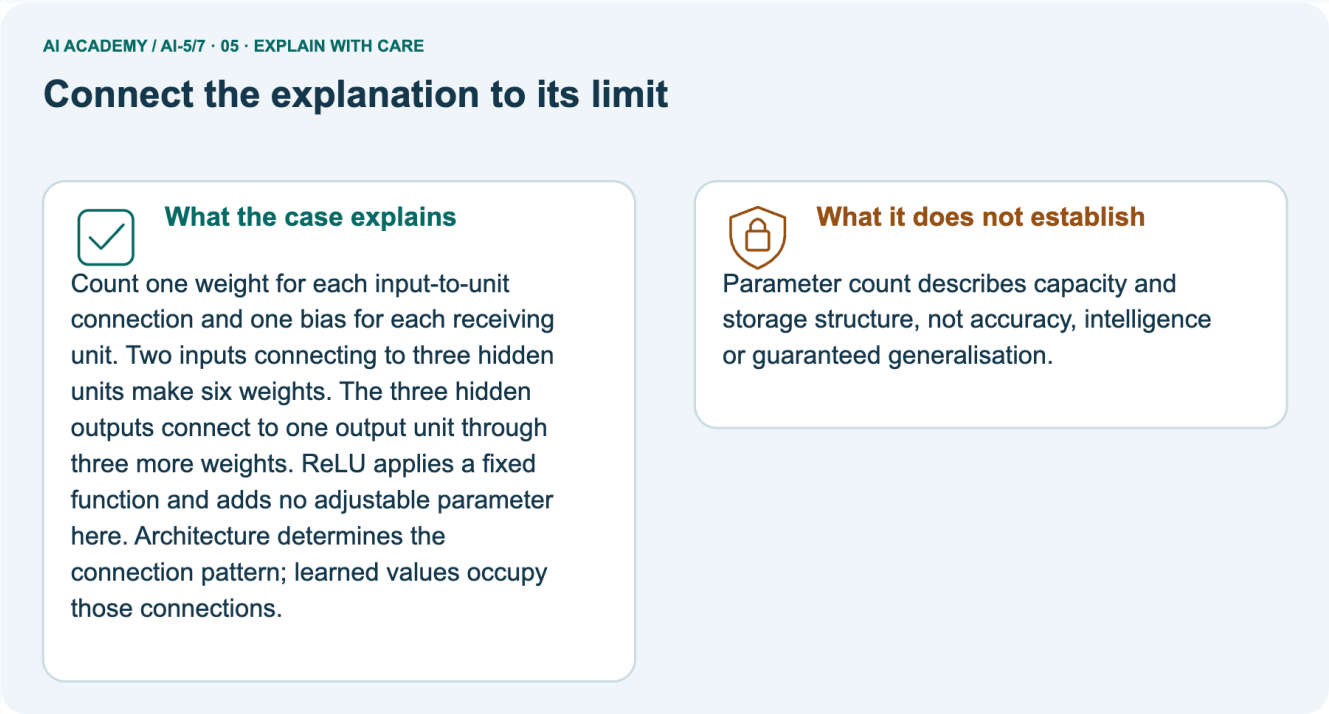
Original case: Total trainable **parameter**s=9+4=13; ReLU adds no **parameter**s here. Changed condition: Increase hidden width from 3 to4. Expected result: First layer 12 **parameter**s; **output** layer 5; total 17.

Reason: Each new hidden unit adds 2 **input** weights, 1 hidden bias and 1 outgoing weight.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Count one weight for each **input**-to-unit connection and one bias for each receiving unit. Two **input**s connecting to three hidden units make six weights.

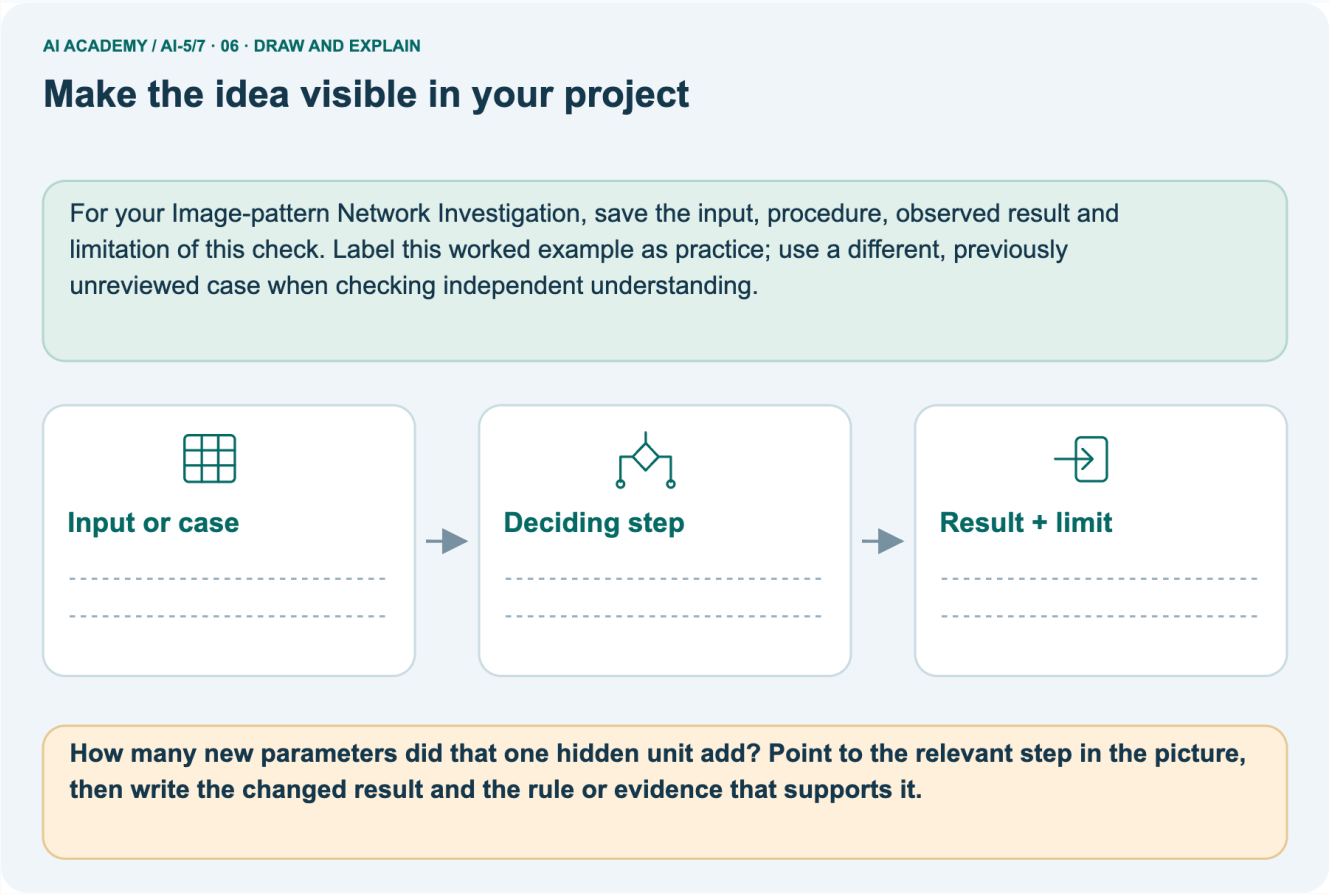
The three hidden **output**s connect to one **output** unit through three more weights. ReLU applies a fixed function and adds no adjustable **parameter** here.

Architecture determines the connection pattern; learned values occupy those connections. Limit: **Parameter** count describes capacity and storage structure, not accuracy, intelligence or guaranteed generalisation.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: How many new **parameter**s did that one hidden unit add? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Parameters and architecture: the case

02. Follow the deciding steps

03. Trainable parameter count

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.